CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

Forth Semester of B. Tech. (CL) Examination
May 2011

CL 210 Hydrology and Hydraulics

Date: 12.05.2011, Thursday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Explain with neat sketch the hydrological cycle showing all it's components. **[2]**

(b) Discuss the types of hydrological data and it's sources. **[3]**

(c) Explain with neat sketch 'Pan evaporimeter'. **[2]**

Q - 2 (a) Define: Hydrograph and Hyetograph. Explain Electrical resistivity method of ground water investigation in detail. **[4]**

(b) A sub-basin has six numbers of rain gauges. The annual rainfall recorded by gauges is given below. Considering 8% error in the estimation of mean annual rainfall. Calculate optimum number of gauges required for the sub-basin. **[5]**

Rainguage Station	A	B	C	D	E	F
Mean Annual Rainfall (cm)	102	77	84	53	66	80

(c) Estimate the maximum flood flow for the following catchments by using an appropriate empirical formula: **[5]**

(i) $A_1 = 35.5$ km^2 for Western Ghat area , Maharashtra

(ii) $A_2 = 30.5$ km^2 in Gangetic plain, where, $C_D = 6$

What is the peak discharge for $A_3 = 40.0$ km^2 by maximum world-flood experience?

OR

Q - 2 (a) What are the factors affecting runoff? Discuss well losses in detail. **[4]**

(b) Areas of Theisson's polygons and precipitation values are tabulated for a drainage basin of area 1000 km^2. Estimate the total volume of the rainfall received in m^3 over a basin. **[5]**

Station	A	B	C	D	E	F	G	H
Theisson's Polygon Area (km^2)	166	170	151	157	121	33	125	45
Precipitation (cm)	11	9	11	12.2	14	15	14.3	12.9

(c) Define flood and flood routing. Describe any one method of flood routing in detail. **[5]**

Q - 3 (a) The time-drawdown data from an observation well at Bhil located in Gujarat are given below. The test well is pumped at rate of 2000 lpm. Draw a drawdown graph 's' v/s '(t / r^2)', where, r = 120 m. Also determine the aquifer constants 'T' and 'S' by Jacob's method. **[10]**

Time – Drawdown Data:

Time since Pumping Started 't' (min)	**1**	**2**	**3**	**4**	**6**	**8**	**10**
Observation Drawdown 's' (m)	0.049	0.116	0.161	0.204	0.265	0.302	0.342
Time since Pumping Started 't' (min)	**14**	**18**	**24**	**30**	**40**	**50**	**60**
Observation Drawdown 's' (m)	0.396	0.435	0.482	0.518	0.574	0.610	0.643
Time since Pumping Started 't' (min)	**80**	**100**	**120**	**150**	**180**	**210**	**240**
Observation Drawdown 's' (m)	0.683	0.726	0.760	0.799	0.830	0.857	0.878

(b) Briefly explain the steady radial flow in a well situated in a confined aquifer. **[2]**

(c) An aquifer has 30 m thickness, 12% porosity and bulk modulus of compression, $Es=10^9$ N/m^2. Estimate the storage coefficient of the aquifer. Take bulk modulus of elasticity of the water, $K_w = 2.1$ GN/m^2. **[2]**

OR

Q - 3 (a) Enlist various sub-surface techniques of ground water investigation. Discuss any two techniques in detail. **[7]**

(b) A 20 cm well fully penetrates a confined aquifer 25 m deep. After a long period of pumping at a rate of 1100 lpm, the drawdown in the wells at 25 m and 45 m from the pumping well are found to be 2.5 m and 1.5 m, respectively. Determine the transmissibility of the aquifer. What is the drawdown in the pumped well? **[5]**

(c) Explain following terms briefly: **[2]**

(i) Leakage Factor (ii) Drainage Factor

SECTION – II

Q - 4 (a) Derive the Gyben-Herzberg relation for salt water-fresh water interface. **[5]**

(b) What are the needs of ground water recharge? **[2]**

Q - 5 (a) Describe the components of Roof Rain Water Harvesting System with neat sketch. **[5]**

(b) Which are the factors affecting ground water pollution? Discuss in detail the sources of urban pollution. **[7]**

(c) Explain the phenomena of 'Upconing of saline water'. **[2]**

OR

Q - 5 (a) Write down the approximate demand of water for rural water supply, live stock and industry. [5]

(b) Discuss the sources and nature of pollution from Sewage and Nuclear wastes. [5]

(c) Which preventive measures should be been taken to control the salt water intrusion? [2]

(d) Discuss briefly the pollution from wells and salt water intrusion. [2]

Q - 6 (a) What is ground water pollution? Discuss the pollution mechanism with neat sketch. [4]

(b) State and derive the Darcy's Law for movement of ground water. [6]

(c) Explain the laboratory methods of measurement of permeability in detail. [4]

OR

Q - 6 (a) Discuss the sources and nature of pollution from Industrial and Agricultural wastes. [4]

(b) An aquifer of aerial extent of 126 km^2 is overlain by three strata as given below: [6]

Strata	Thickness (m)	Kx (m/day)	Ky (m/day)
1 (Top)	3	2.5	0.35
2 (Middle)	3	1.5	0.72
3 (Bottom)	1	0.05	0.054

(i) If a 3-hr storm occurs producing a total rainfall 125 mm, estimate the recharge into the aquifer, assuming that the piezometric surface in the aquifer is at the bottom of layer and that all layers are saturated.

(ii) If all three layers are underlain by an impermeable strata instead of the aquifer, estimate the lateral flow per unit width through layers, assuming that the layer dip by 0.25 %.

(c) Define Conjunctive Use. What are the needs of conjunctive use? [4]

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